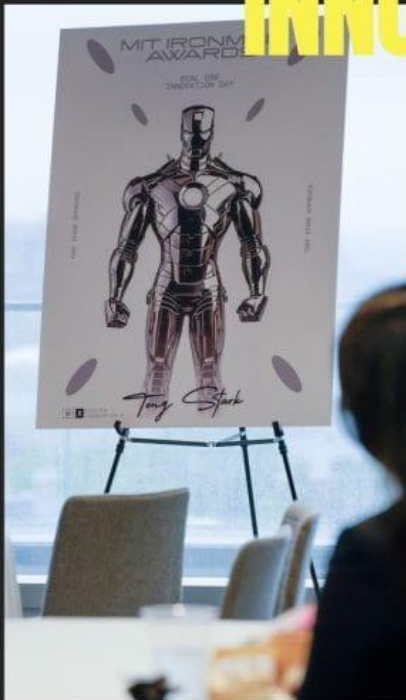




DUAL-USE INNOVATION DAY



EVENT PARTNERS + SPONSORS :

GovSignals

PICOGRID

✦ NOMINAL



Deepnight



MIT HAUS

LINCOLN LABORATORY
MASSACHUSETTS INSTITUTE OF TECHNOLOGY

CHECK-IN + COFFEE

09.00 - 09.30 Towards Waterless Recycling at the Edge

09.30 - 10.15 Edge Manufacturing of Boats, Barracks, Bridges, and Other Structures

BREAK + HARDWARE PETTING ZOO

10.30 - 11.00 Mlx Nominal

11.00 - 11.20 Mlx Picogrid HAC

11.20 - 11.30 Mlx Deepnight

11.30 - 12.00 Mlx Picogrid AID

LUNCH + HARDWARE PETTING ZOO

12.30 - 13.00 MIT Keynote by Prof. Eric Evans

13.00 - 13.40 15.362 Ghostwave

13.40 - 14.20 15.362 Sturgeon

BIO BREAK

14.25 - 15.05 15.362 Strategem Dropworks

15.05 - 15.45 15.362 End Point Assurance

BREAK + HARDWARE PETTING ZOO

16.00 - 16.15 USSOCOM Keynote

16.15 - 16.25 Iron Man Award

16.25 - 16.30 Closing Remarks

MAY 08 2026

SAMBERG CONFERENCE

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DR, CAMBRIDGE,
MA 02142

MIX MISSION INNOVATION X

Customized for MIT 2026 Dual Use Innovation Day
Supervised by Dr AJ Perez



FEATURING THE 2026 USSOCOM-MIT IRON MAN AWARD

Endpoint Assurance

MANTIS

Multilayered Anti-tamper Network for Trusted Integrated Systems



ENDPOINT ASSURANCE

Executive Summary

Endpoint Assurance MANTIS

Lightweight, secure detection of shipment tampering in high-risk supply chain networks

Problem Background

- Shipments pass through third-party, **limited visibility** networks
- Organizations **cannot confirm tampering occurrence** in transit
- Creates **operational and security risk**
- Collab between MIT & U.S. government

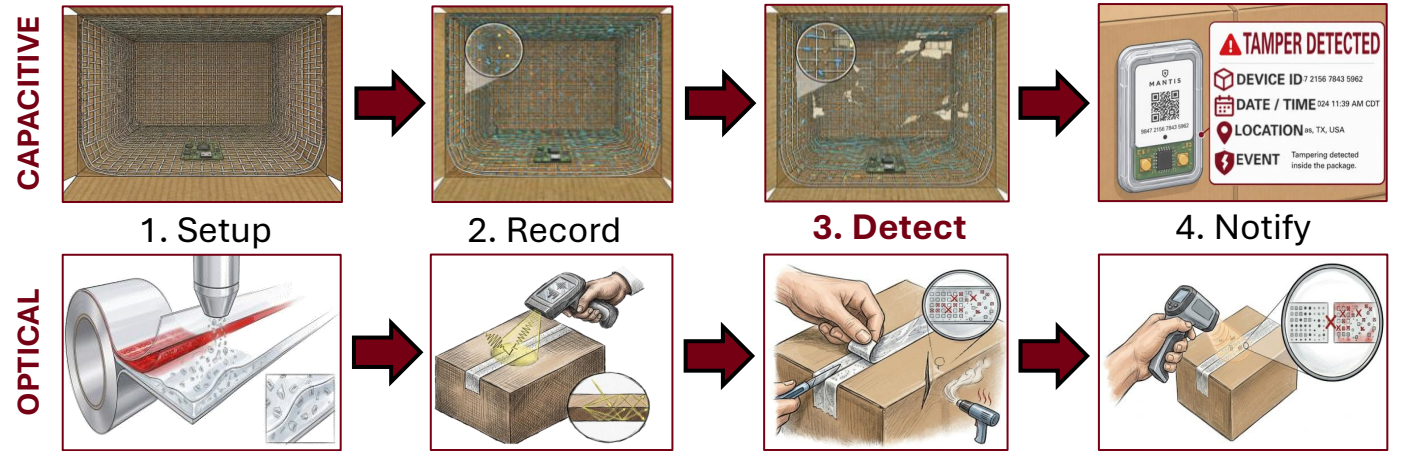
Solution

Multilayered Anti-tamper Network for Trusted Integrated Systems

- Multilayered, lightweight solution to detect **if and when tampering occurs**
- Utilize physical unclonable function technology to **prevent spoofing**

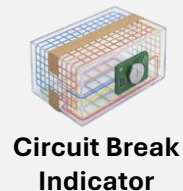
Next Steps

- Securing **funding and scale-up partners**



CORE TECH

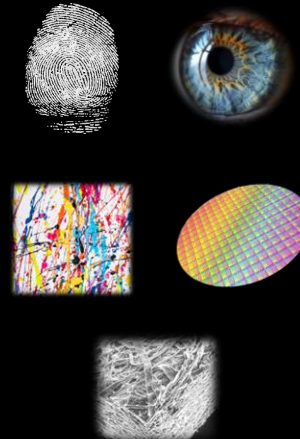
Multi-Layered Solution



- Provide redundancy
- Identify if + when

PUFs

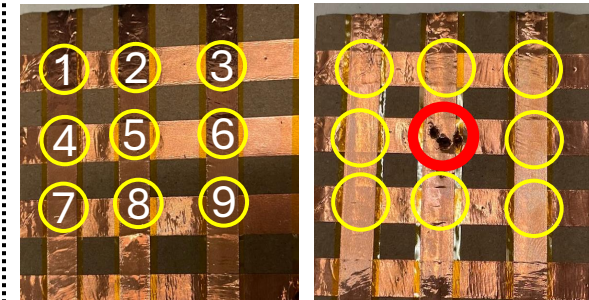
Physical Unclonable Functions



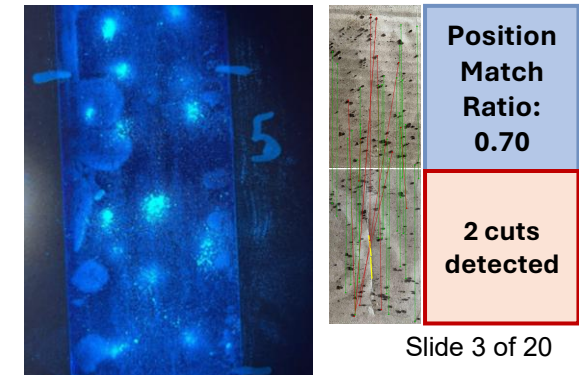
- Unique identifiers
- Prevent spoofing

PROTOTYPING

CAPACITIVE



OPTICAL



Team: Endpoint Assurance

Executive Leadership



Mike Fischer
Team Lead



Greta Reitenbach
Chief of Staff



Engineering



Sydney Saenz
Engineering Lead



Corey Decker
Engineering Team



Garrett Campagna
Engineering Team



Josephine Lee
Engineering Team



Integration & Ops



Kenan Sehnawi
Integration Lead



Brendan Owen
Integration Team



Chandler Smith
Integration Team



Evan Hoch
Integration Team



Cam Heard
Integration Team



Business & Strategy



Aidan Kenealy
Strategy Lead



Alfred Wong
Strategy Team



Amit Govrin
Strategy Team



Josh Cohen
Strategy Team



Sam Yallowitz
Strategy Team



Fritz Demopolous
Strategy Team



The Problem: Undetectable Tampering in Transit

What We Do

We **detect parcel tampering** through opaque, third-party logistics networks.

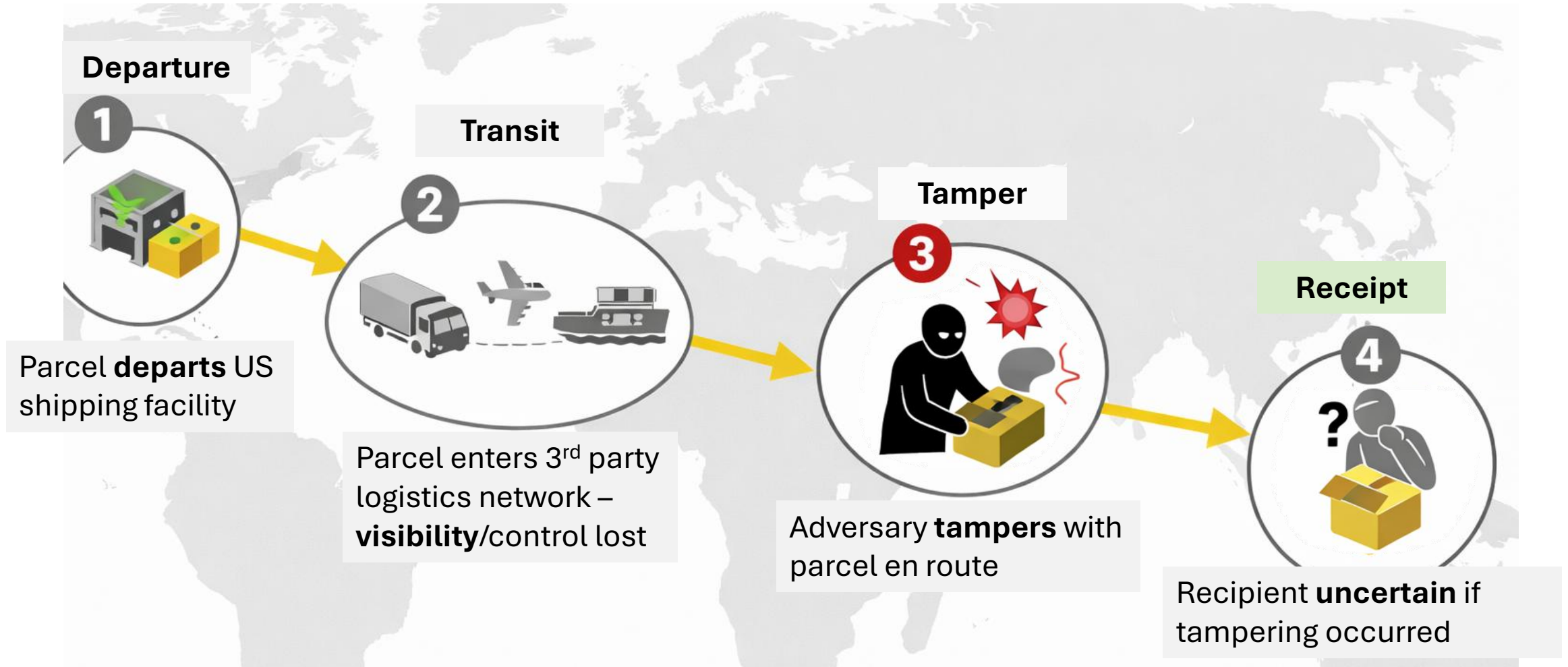
Why It Matters

- Shipments pass through third-party networks with **no visibility**.
- Organizations **cannot confirm tampering** in transit.
- This creates **operational and security risk**.

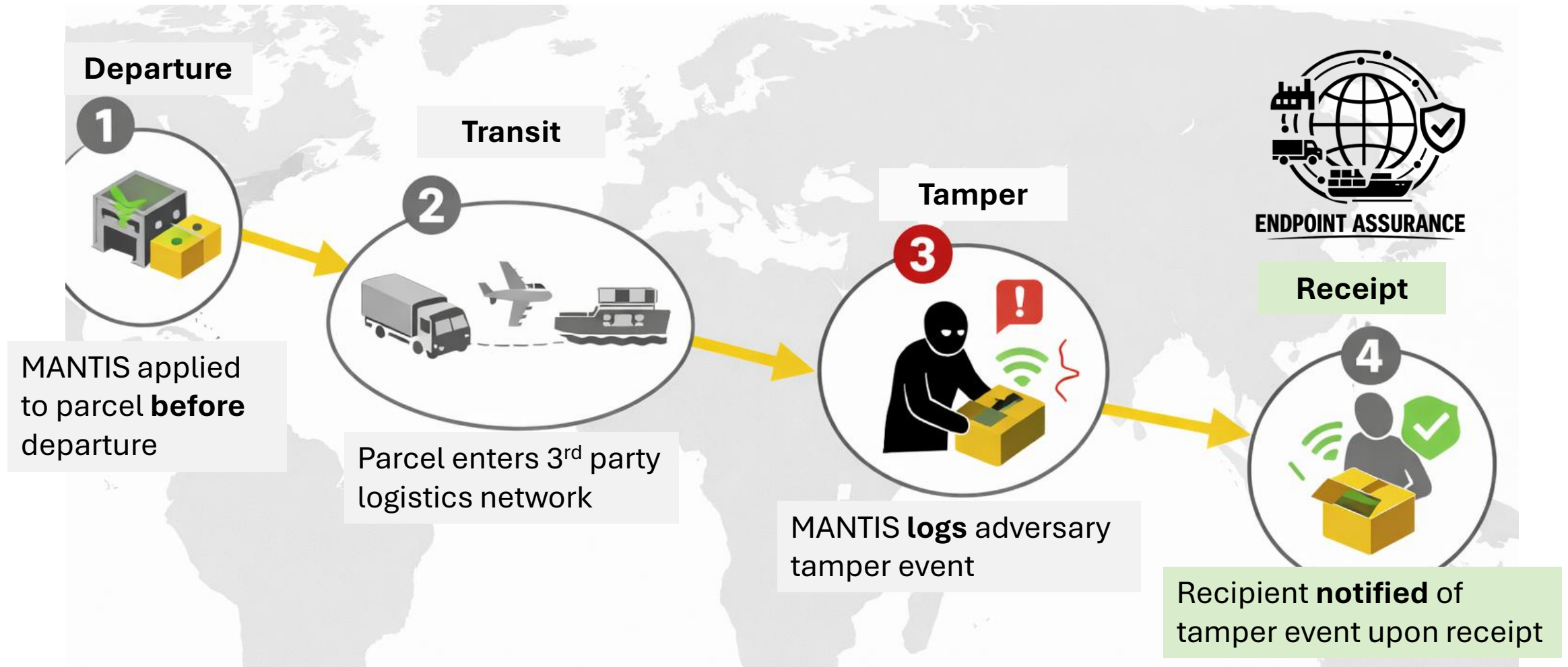
Design Constraints

- No external devices on parcels.
- Works across air, sea, and ground shipping.
- Low cost and scalable.

Status Quo: Uncertainty About Tampering During Shipment



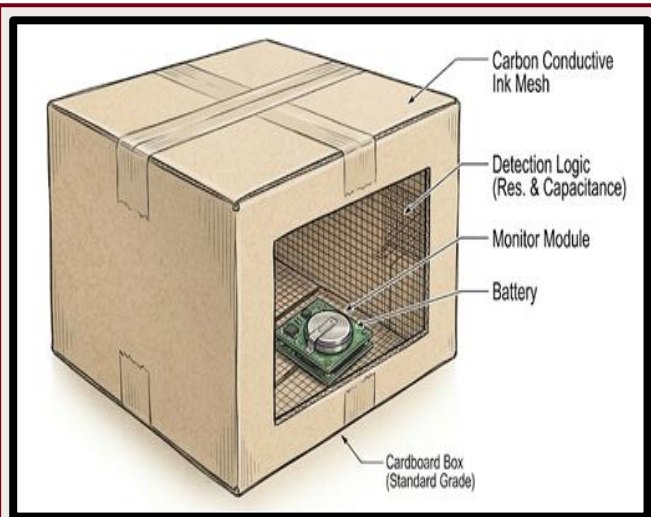
Endpoint Assurance: Certainty About Tampering During Shipment



Customers want to **catch tampering when it happens**; if that is not possible, they want to identify that tampering occurred.

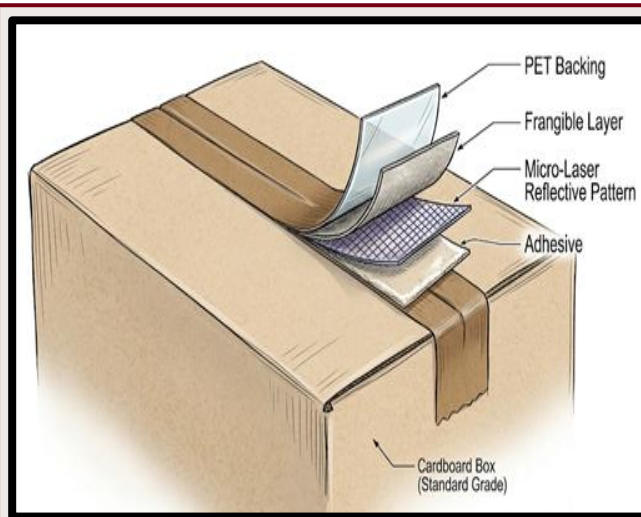
Endpoint Assurance: Overview of MANTIS

The combined solution of **engineered tamper tape** and an **embedded circuit-break indicator** was chosen based on its ability to provide enhanced security with dual detection capabilities.



Internal Circuit-Break Indicator

- Carbon conductive ink disguises traces from X-ray inspection
- Separate monitor module wirelessly queries circuit integrity



External Engineered Tamper Tape

- Standard packing tape appearance
- Micro-laser IR-reflective security pattern
- Textured backing for tamper detection



MANTIS



Dual-layer redundancy



Tamper log with timestamp



Spoofing-resistant

Assurance at every endpoint.

Explainer: Physical Unclonable Function (PUF)

The solution uses a physical unclonable function (PUF) to exploit random physical variations, thus reducing the risk of cloning and spoofing.



THREAT: Adversaries with advanced manufacturing capabilities can bypass, clone, or spoof traditional anti-tamper seals and tracking tags.



COUNTERMEASURE: We introduce a *physical unclonable function (PUF)* integrated directly into cargo packaging, monitored by machine learning baseline.



ADVANTAGE: PUFs are *constant* and *unique* to an object. Adversary with access to all knowledge on how PUF was created would still not be able to replicate it.

WHAT IS A 'PUF'?

- ✓ A physical “fingerprint” created by the inherent, microscopic chaos of a manufacturing process
- ✓ Statistically impossible to replicate, even by original manufacturer

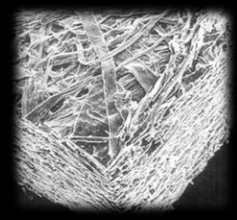
Real world PUF examples



Fingerprints



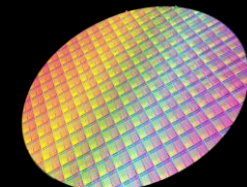
Iris



Paper fibers



Paint splatters

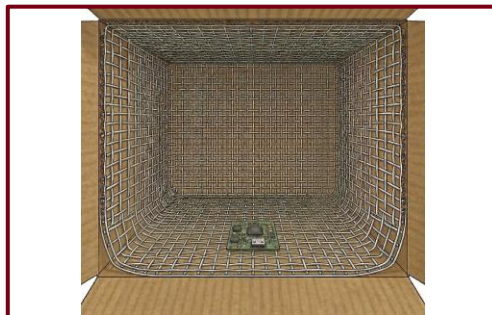


Silicon wafers

Capacitive PUF Circuit Break Indicator Overview

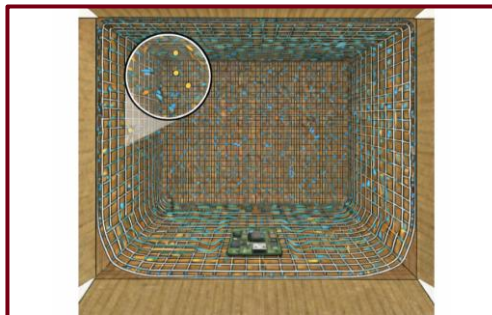
HOW IT WORKS

1



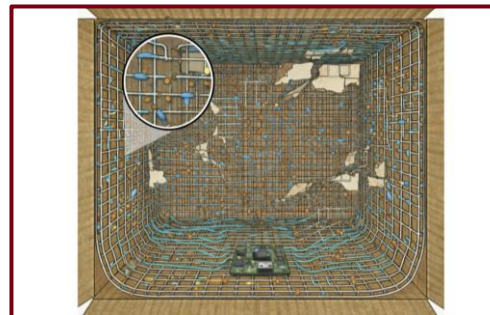
Carbon-conductive mesh lines inside of box

2



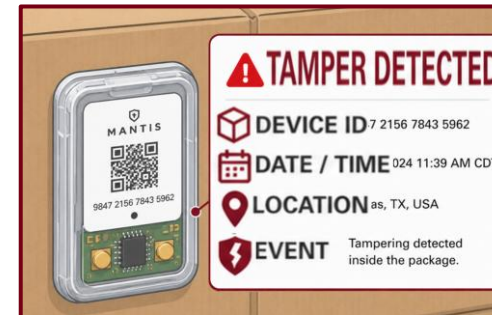
Dielectric particles cure, setting unique capacitance

3



Physical tampering alters capacitance

4



Tamper event logged with timestamp



360° Protection

Offers high-sensitivity protection against adversary tampering from any angle.



Unique Capacitive Fingerprint

Variations ensure that replicating boxes will still register different capacitance.



Dedicated ASIC Monitor

Cryptographically protected to ensure data integrity.



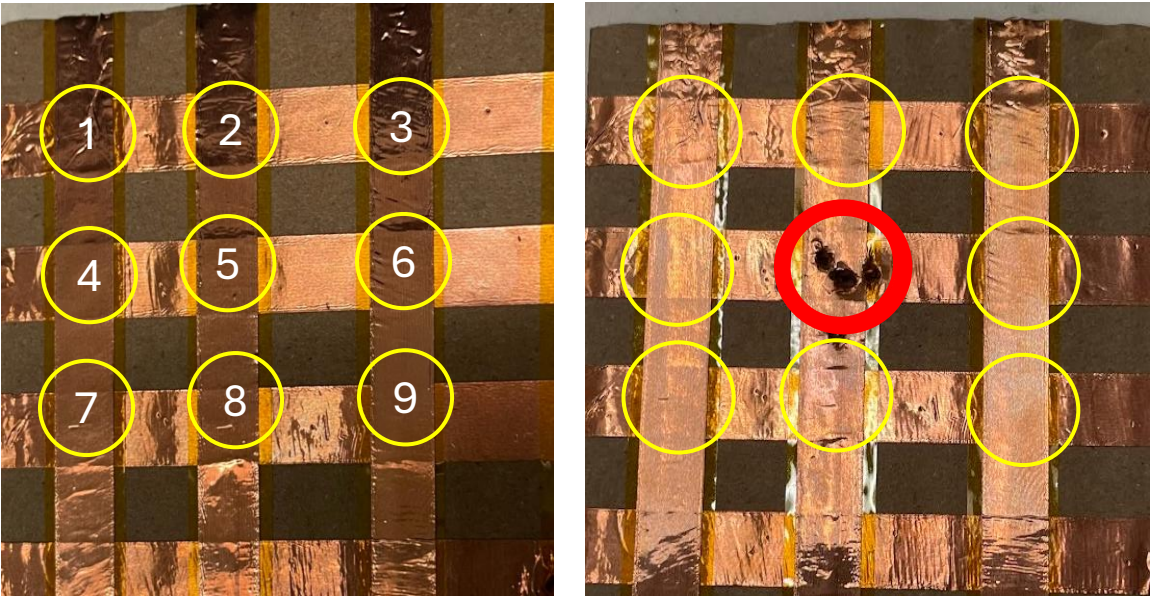
Layered Defense

Multi-layered detection architecture, with discrete & complimentary components.

WHY IT WORKS

Capacitive PUF Prototype Demonstration

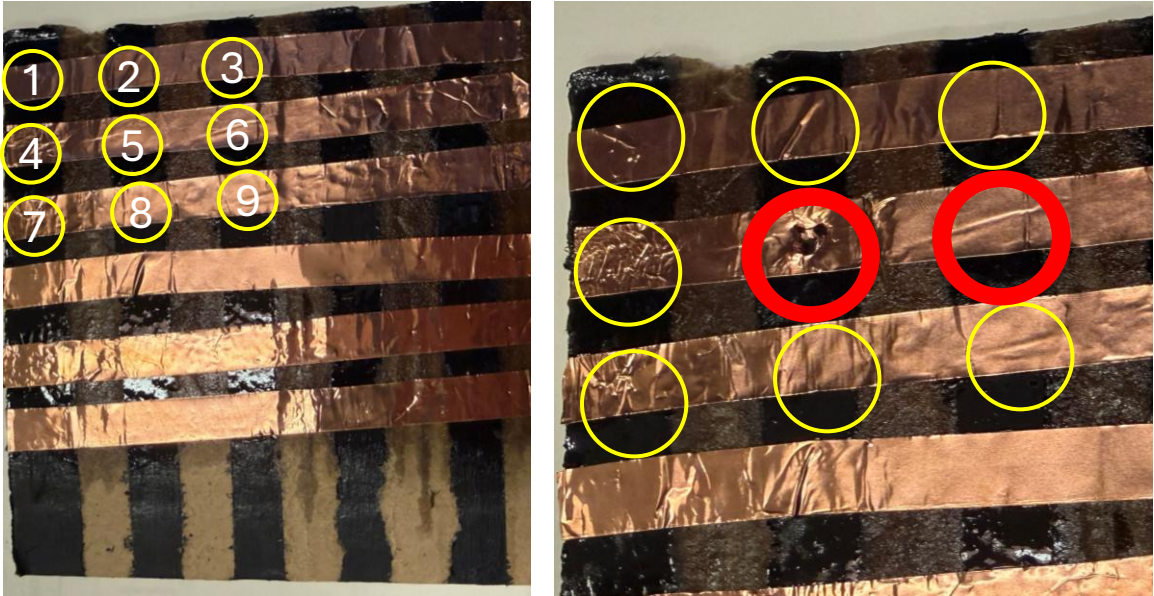
Array 1: Copper Tape



Before Node 5 Tampered After

Node	1	2	3	4	5	6	7	8	9
% Change	+2.4	+7.4	+0.5	+43.4	-2.3	+47.7	+1.5	+8.8	+1.6

Array 2: Carbon Conductive Paint



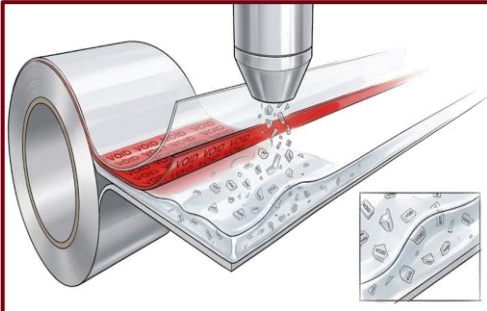
Before Node 5/6 Tampered After

Node	1	2	3	4	5	6	7	8	9
% Change	+5.1	-8.4	-4.0	-14.9	-9.1	-51.3	-8.4	-0.8	+42.1

Optical PUF Tamper Tape Overview

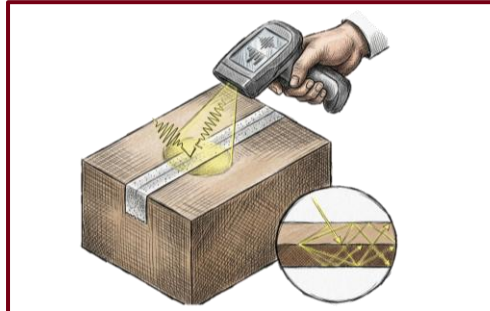
HOW IT WORKS

1



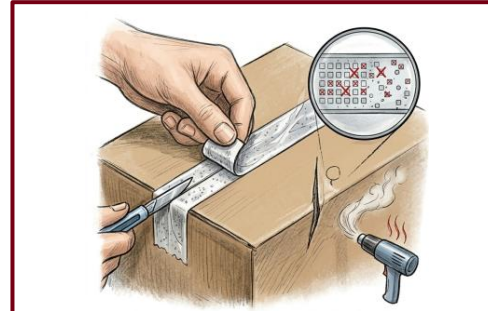
IR-reflective particles
embedded in tape

2



Pattern imaged and
recorded at origin

3



Tampering alters particle
distribution

4



Computer vision flags
potential tampering



Stochastic Particle Distribution

Micrometer-scale IR
grain positions are
mechanically random
and unreproducible.



Frangible Substrate

Kraft base delaminates
irreversibly — peel-and-
replace is physically
impossible.



Substitution Defense

Same-roll replacement
tape always produces a
mismatching stochastic
pattern.



Classical CV Authentication

SIFT + Canny edge diff:
interpretable, auditable
evidence — no neural
net required.

WHY IT WORKS

Initial PUF Exploration | Coffee Grounds and Packing Tape

Classical computer vision pipeline

- No neural network, model training, or GPU required
- Deterministic and highly stable

Lightweight & highly scalable

- Runs in ~60 sec per batch of 10
- Easily integrates into more complex models

Detects ground movement + cuts

- SIFT matching & edge differencing
- Validated on 20 tamper/untampered image pairs

Machine Learning Model

**Original seal
(untampered)**



2

**Thermal drift
(ambient heat)**



2

**Mechanical
tampering**



2

No tampering

Tampering

ML model output – difference map and cut detection

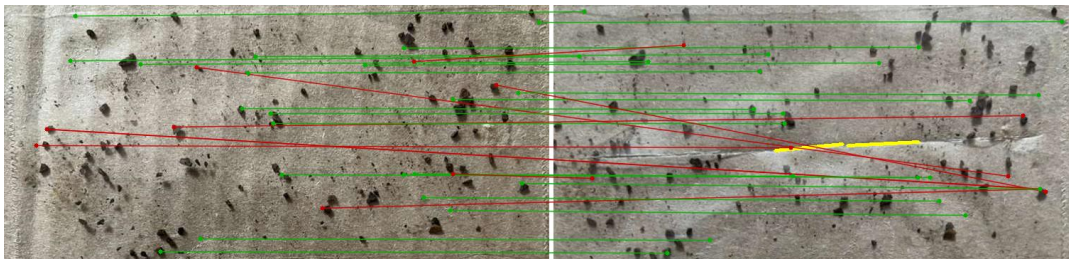
Untampered



Tampered



SIFT matching + cut detection



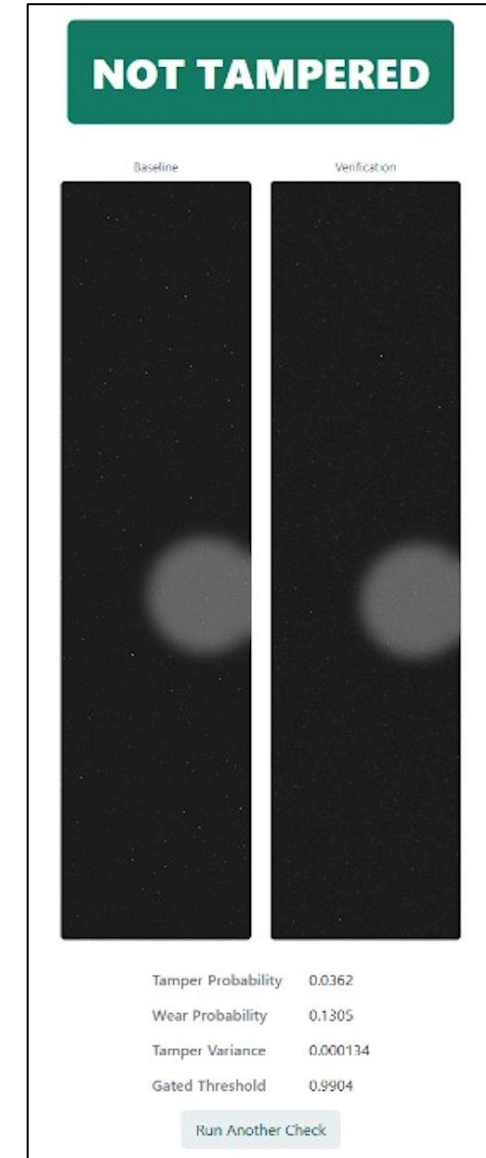
Position Match Ratio: 0.70

2 cuts detected

Synthetic Proof of Concept

Siamese deep network trained on simulated data

- 1,000 optical PUF images simulated, with synthetic perturbations mimicking benign rescans, wear, and tampering.
- Deep Siamese network compares enrollment vs. verification: shared encoders, geometric alignment, and fused features drive separate tamper and wear detection heads.
- Outputs tamper and wear scores; thresholded for low false positive rate.

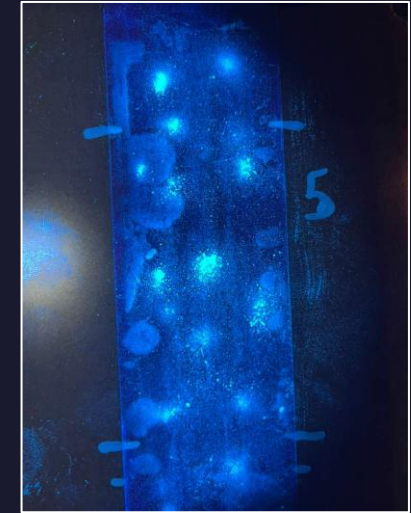


Example run from synthetic model

UV PUF Prototype Demonstration

1 Apply and Scan Optical PUF Seal

- UV-fluorescent particle tape applied to surface
- Random distribution creates a unique, irreproducible fingerprint per seal
- Seal imaged under UV at application and delivery



PUF Seal Applied to Surface

(UV-fluorescent particles in tape)

UV PUF Prototype Demonstration

2 Analyze Image Pair

- Image scans aligned, reprojected, and analyzed for pattern matches
- Metrics extracted via SIFT feature matching, homography alignment, and similarity scoring (SSIM, histogram distance, NCC)
- Metrics fed into a gradient boosting ensemble (optimized for small datasets)
- **Output:** continuous tamper score or binary tamper / no-tamper classification

Achieves accuracy of 85% with 0% False Positive Rate

Sample 1 – No Tampering

Feature Matching

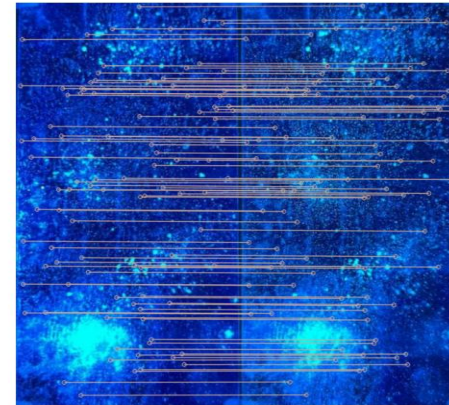


Photo 1

Photo 2

Tamper Likelihood
Score

0.83

Model Output

NO TAMPER DETECTED

Threshold: 0.859

Sample 2 – Tampering: Peel and Replace

Feature Matching

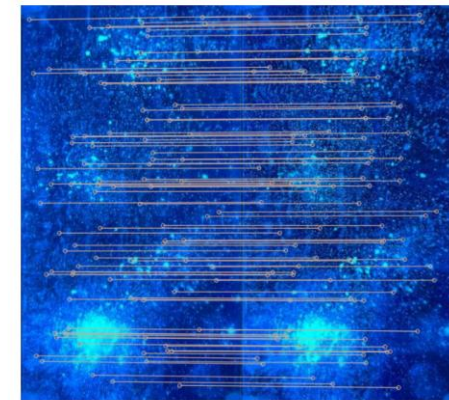


Photo 1

Photo 2

Tamper Likelihood
Score

0.860

Model Output

TAMPER DETECTED

Threshold: 0.859

UV Tamper Tape Model Analysis Demonstration

Deploy

UV Tape Tamper Checker

Select two UV images from the dataset (or upload), run the model, and review a clean tamper risk summary.

Image Source

☐ Select from repository ☒ Upload images

Image 1

Upload

200MB per file • JPG, PNG

Image 2

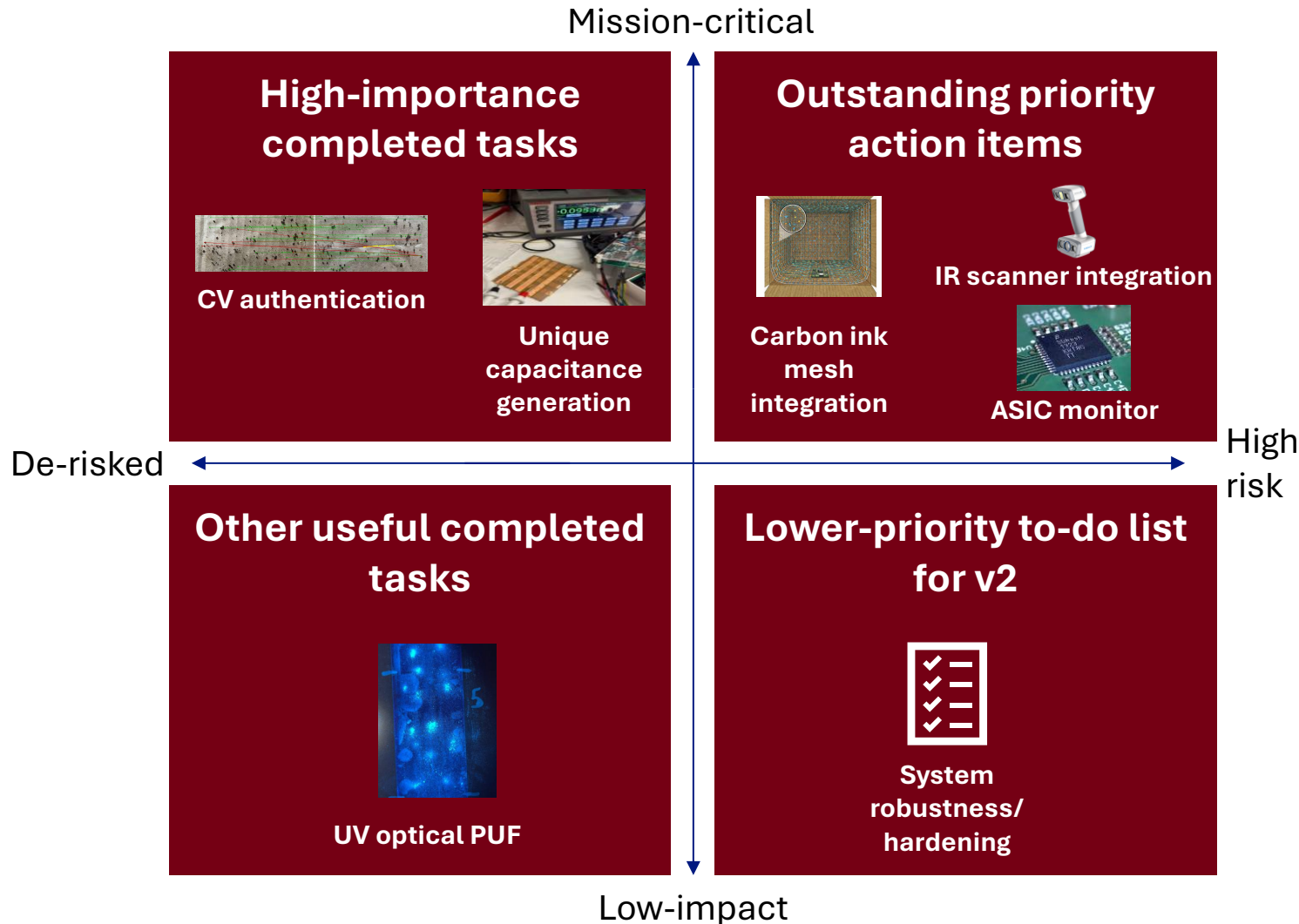
Upload

200MB per file • JPG, PNG

Run Tamper Analysis

Next Steps: Risk/Impact Assessment and Way Ahead

Risk/Impact Assessment



Proposed Forward Plan

T+30 days

- IR scanner integration
- Carbon ink mesh validation
- ASIC monitor
- Delivery of integrated alpha prototype

T+90 days

- Prototype field testing and iteration
- Draft documentation:
 - Threat Vulnerability Assessment
 - CONOPS

T+180 days

- Full-scale chain of custody demonstration
- Sophisticated adversary testing
- Subsystem hardening
- Develop scalable production process
- Pre-seed raise and SBIR application

MANTIS turns a ~\$20B Annual Logistics Problem into a Scalable Dual-Use Solution

2026 Status Quo

250B packages
0.01% "high value" = 25M pkgs.
0.4% tamper rate = 100K pkgs.
~\$202K per high-value tamper =
\$20.2B annual loss



Value Proposition

\$20.2B annual loss
10% WTP for package assurance
\$2B WTP-adjusted market
1% capture assumption =
\$20M annual revenue

Scaling Flywheel Enables Core Mission

1. Initial USGA validation and performance credibility
2. Adoption by aero, electronics, pharma, and luxury
3. Commercial volume expands production capacity
4. Lower unit cost and stronger USGA/DoW readiness

MANTIS Architecture Enables Product Tiers

- **Full MANTIS System:** Highest-security missions
 - **MANTIS Mesh:** Detects covert penetration
 - **MANTIS Tape:** Consumable seam security
- Full-stack assurance **and** risk-tiered segmentation*

Executive Summary

Endpoint Assurance MANTIS

Lightweight, secure detection of shipment tampering in high-risk supply chain networks

Problem Background

- Shipments pass through third-party, **limited visibility** networks
- Organizations **cannot confirm tampering occurrence** in transit
- Creates **operational and security risk**
- Collab between MIT & U.S. government

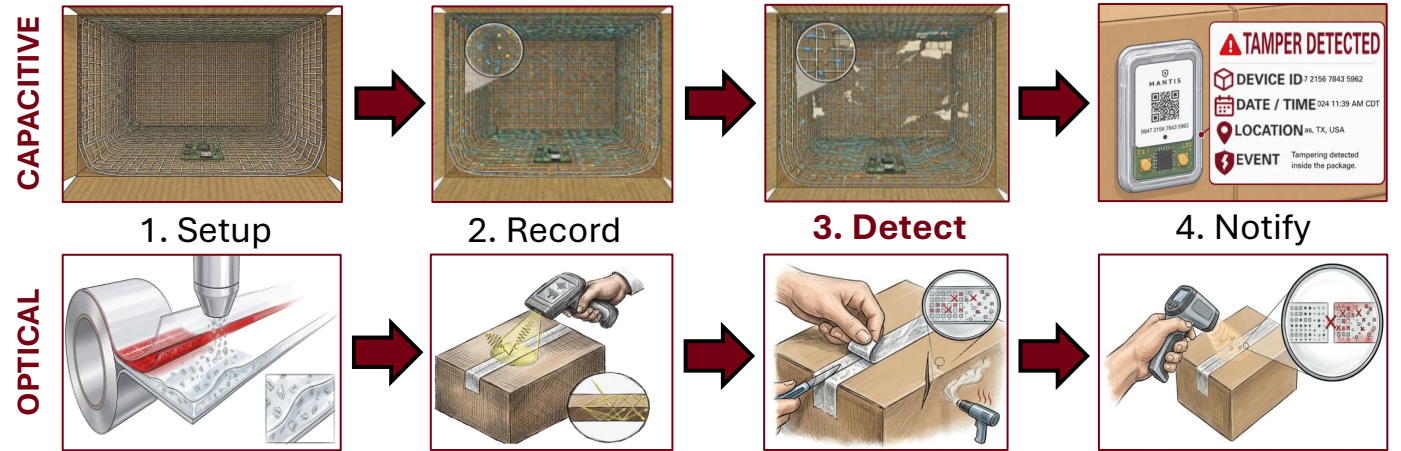
Solution

Multilayered Anti-tamper Network for Trusted Integrated Systems

- Multilayered, lightweight solution to detect **if and when tampering occurs**
- Utilize physical unclonable function technology to **prevent spoofing**

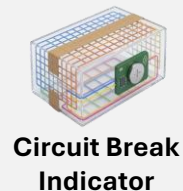
Next Steps

- Securing **funding and scale-up partners**



CORE TECH

Multi-Layered Solution



Circuit Break Indicator

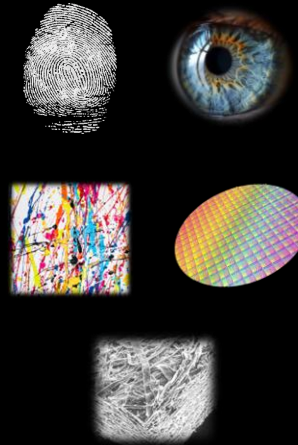


Low-Signature Tamper Tape

- Provide redundancy
- Identify if + when

PUFs

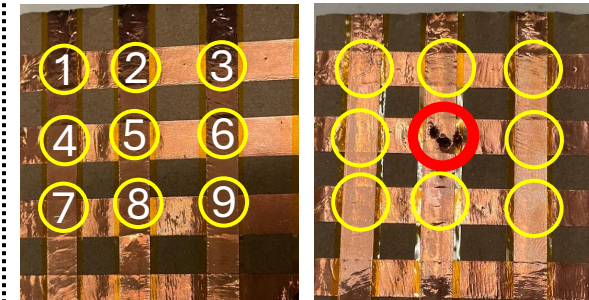
Physical Unclonable Functions



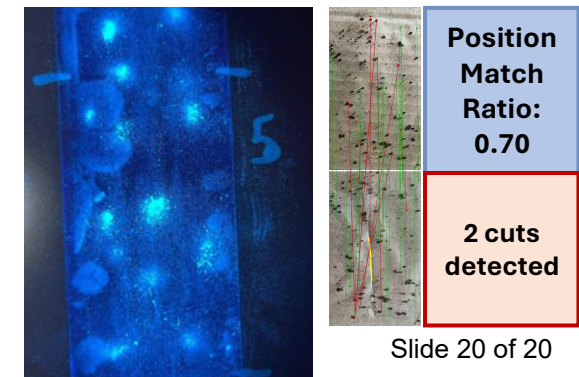
- Unique identifiers
- Prevent spoofing

PROTOTYPING

CAPACITIVE



OPTICAL



APPENDIX

Optical PUF Tamper Tape Tech Deep Dive

HOW IT WORKS

1

Manufacture

IR-reflective particles are deposited stochastically into tape adhesive. Each length of tape carries a unique, irreplicable particle distribution.

2

Enrollment (Sender)

UV/IR scanner captures the baseline image of the applied tape at origin. The pattern is transmitted securely to recipient via Endpoint Assurance app.

3

Transit

Frangible kraft substrate delaminates on any tamper attempt. The particle field is permanently disturbed, making reapplication is impossible.

4

Verification (Recipient)

Receipt scan is compared to baseline via SIFT keypoint matching + Canny edge differencing. Mismatch results in tampering flagged.

WHY IT CANNOT BE CLONED



Stochastic Particle Deposition

IR-reflective grain positions are mechanically random at the micrometer scale meaning even the original manufacturer cannot reproduce the same pattern twice.



Frangible Substrate

Kraft-paper base delaminates irreversibly on removal, permanently destroying the particle field. Peel-and-replace attacks are physically impossible.



Substitution Defense

Any fresh tape from the same roll produces a different stochastic pattern. The sender's transmitted baseline will always mismatch a replacement seal.

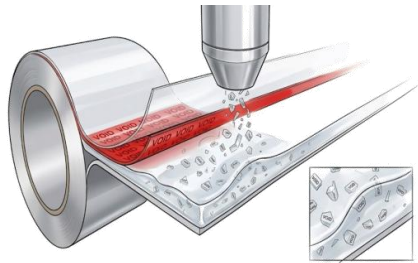


Classical CV Authentication

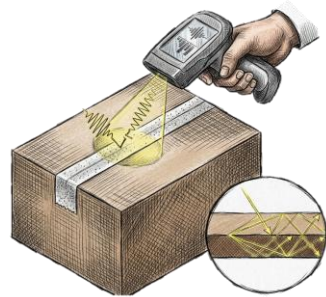
SIFT keypoint matching + Canny edge differencing provides interpretable, auditable tamper evidence with no black-box neural net required.

Multi-Layered PUF Tamper Indication System

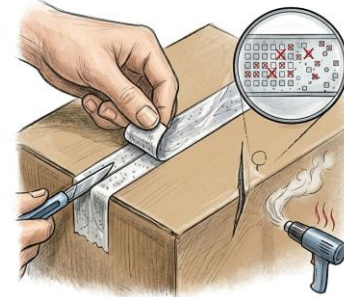
1 Optical PUF (Tamper Tape)



IR-reflective particles
embedded in tape



Pattern imaged and
recorded at origin

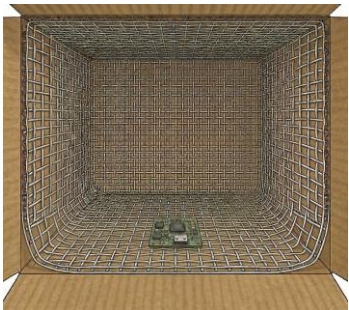


Tampering alters
particle distribution

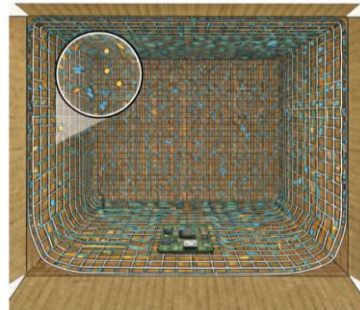


Machine learnings flags
as potential tampering

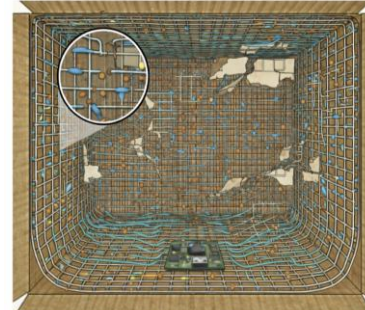
2 Capacitive PUF (Circuit Break Indicator)



Carbon-conductive
mesh lines inside of box



Dielectric particles cure
randomly, setting
unique capacitance



Physical tampering
alters capacitance



Tamper event logged
with timestamp

Sponsor Feedback on Top 4 Concepts

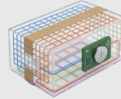
The sponsor provided us with their ranked list of concepts from our top 4 concepts, prioritizing a design that combines the external low-signature tamper taper with an embedded circuit-breaker indicator

Sponsor ranked list of top 4 concepts

1 *Low-Signature Tamper Tape*



2 *Circuit-Break Indicator*



3 *Logger + Anomaly Detection Model*

4 *One-Time Tamper Indicator*

What Does Our Sponsor Want to See?

Key Performance Objectives

- Provide high-confidence indication of tampering
- Minimize false-positive rate
- Minimize adversary detection/defeat likelihood
- Perform within specified environmental conditions

Key Tensions & Tradeoffs

- High-confidence detection vs. false positives
- Scalability vs. technical complexity
- Sensitivity vs. resilience in harsh environments

Prototype component List

Major component	Cost Type	Quantity	Unit Price	Est. Cost	Procurement	Manufacturer
Optical PUF Components						
PET film	Raw material					
Textured backing	Raw material					
Additives						
Frangible primers						
Nickel shim						
UV-curable resin						
High-tack HMPSA						
Chemical “solvent-swell” dyes						
Photochromic ink						
Capacitive PUF Components						
Carbon conductive ink						
Capacitance-to-Digital Converter						Analog Devices AD7147
PET film with dielectric						
Infineon PSoC 5 Microcontroller						
Accelerometer						
Photodiode						
Thin flexible copper tape						
LiPo battery						
Cardboard boxes						

Production component List

Major component	Cost Type	Quantity	Unit Price	Est. Cost	Procurement	Manufacturer
Optical PUF Components						
PET film	Raw material					
Textured backing	Raw material					
Additives						
Frangible primers						
Nickel shim						
UV-curable resin						
High-tack HMPSA						
Chemical “solvent-swell” dyes						
Photochromic ink						
Capacitive PUF Compnents						
Printed conductive material						
Capacitance-to-Digital Converter						Analog Devices AD7147
PET film with dielectric						
ASIC						
Accelerometer						
Photodiode						
Faraday enclosure						
LiPo battery						
Cardboard boxes						

Capacitor Array Testbed Data

Capacitance values of each node before and after tamper event (slice, drill)

Copper Tape Array Data (Tamper Location: Node 5)									
Node #	1	2	3	4	5	6	7	8	9
Untampered Capacitance (pF)	43.85	44.39	43.68	46.21	45.09	44.18	44.44	44.11	44.35
Tampered Capacitance (pF)	44.92	47.69	43.91	66.25	-104,400,000	65.26	45.11	48.01	45.04
Percent Change	+2.4%	+7.4%	+0.5%	+43.4%	-2,315,368.2%	+47.7%	+1.5%	+8.8%	+1.6%

Carbon Conductive Paint Array Data (Tamper Location: Between Nodes 5 & 6)									
Node #	1	2	3	4	5	6	7	8	9
Untampered Capacitance (pF)	235.0	198.2	180.9	160.9	146.8	171.4	169.9	152.9	175.7
Tampered Capacitance (pF)	247.1	181.6	173.7	137.0	133.5	83.5	155.7	151.6	249.6
Percent Change	+5.1%	-8.4%	-4.0%	-14.9%	-9.1%	-51.3%	-8.4%	-0.8%	+42.1%

PUF CV Model: Siamese-Attentional Model Framework

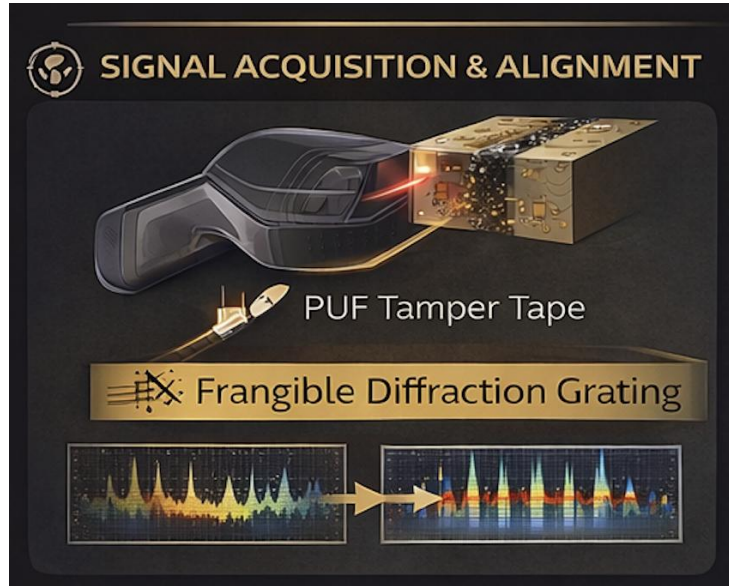
AI architecture for detecting tampering while filtering environmental wear.

System Architecture Overview

5 Core Components

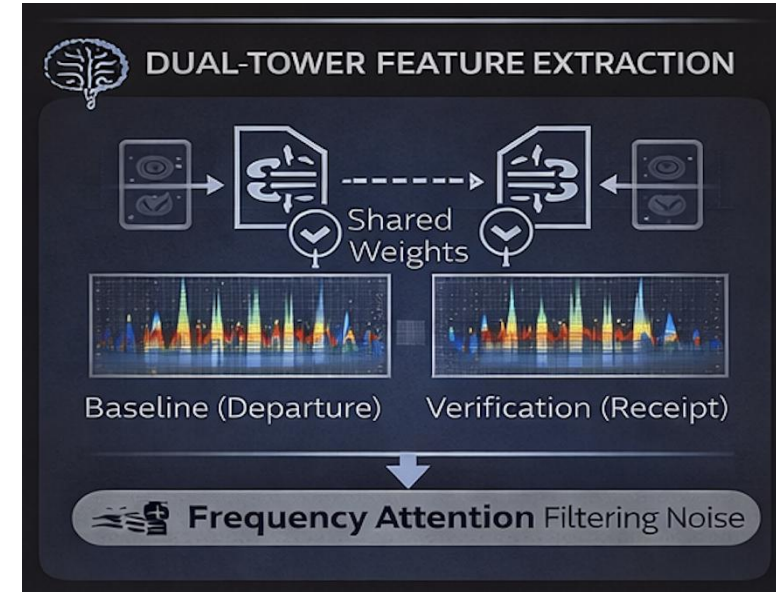
- Signal acquisition & alignment
- Dual tower feature extraction
- Differential change detection
- Wear vs. tamper decision engine
- False positive suppression

Signal Processing & Feature Extraction



Input Processing:

- Dual intensity scans
- Spatial alignment correction using spatial transformer network (STN)
- Z-score signal normalization



Feature Extraction:

- Siamese neural network
- Dilated (Atrous) convolution detection
- Frequency attention filtering

Wear vs. Tamper Decision Engine

Dual Classification Heads:

Tamper model:

- Structural fracture detection
- Security compromise identification

Wear model:

- Environmental degradation detection
- Decision thresholds dynamically adjusted



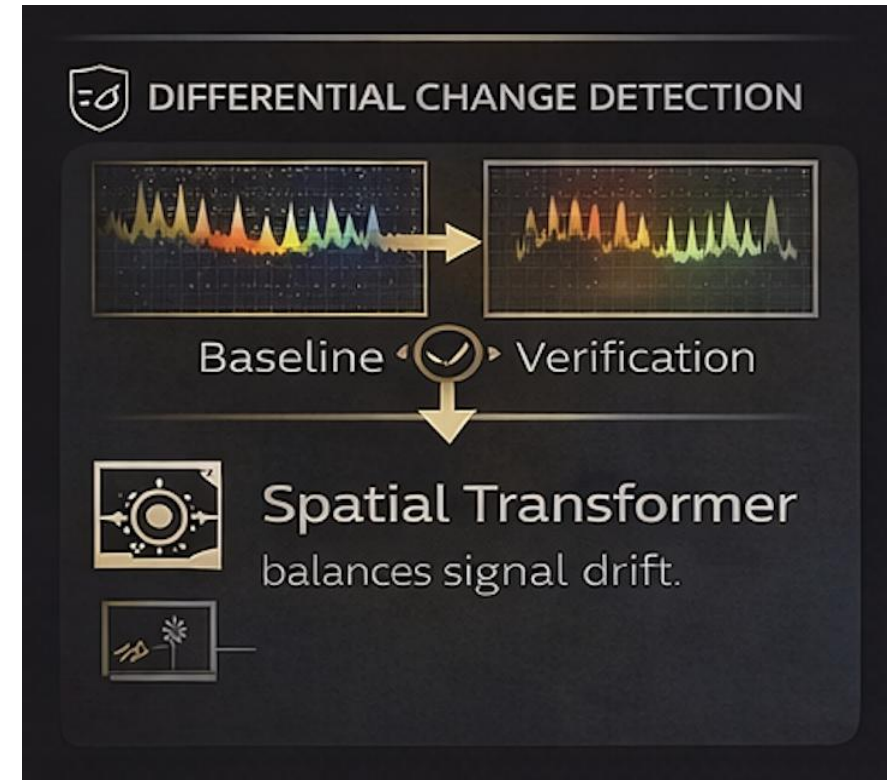
Intelligent Change Detection

Differential latent comparison detects topology changes

Distinguishes:

- True tampering
- Normal wear
- Environmental degradation

Result: Reduced false alarms



False Positive Prevention & Value

Controls:

- 50x false positive penalty
- Bayesian uncertainty detection
- Multi-pass Monte Carlo verification
- $\geq 98\%$ confidence threshold

Value:

- Fewer inspections
- Trusted chain-of-custody
- Reliable logistics security

